CS {4/6}290 & ECE {4/6}100 - Spring 2021

Homework 1: Performance Evaluation

Dr. Thomas Conte

Due: January 29th, 2021 @ 11:59 PM

Rules

- **This is an individual assignment. ABSOLUTELY NO COLLABORATION IS PERMITTED**. All cases of honor code violations will be reported to the Dean of Students.
- The due date at the top of the assignment is final. Late assignments will not be accepted.
- Please use office hours for getting your questions answered. If you are unable to make it to office hours, please email the TAs.
- Make sure you **show your work and give explanations** for all questions unless otherwise stated.
- All answers must be **handwritten or typed**, and an electronic copy (PDF format) uploaded to Canvas before the due date.
- Make sure your answers are written within the boxes we have provided. We will be using Gradescope for grading and answers outside the boxes will be recognized by the system.

Question 0 – Short Answer Questions

1. According to Flynn's Taxonomy of classifying instructions and data that are processed by a machine, what would be the best way to classify the current generation of x86 Intel processors? **Please state your reasoning.**

MIMD – Can run multiple programs one or more per core making the processor a multiple instruction multiple data capable processor

2. Out of the 4 levels of parallelism discussed in lecture which one is the programmer responsible for and **why?**

Thread Level Parallelism – Programmer is responsible for 'parallelizing' the code via algorithmic modifications and (or) use of parallelization libraries.

3. Which of the "2.5 architecture tricks" can adding a Cache to a processor be reduced down to? **Please explain.**

Speculation w/ prediction

A cache speculates that the processor will request memory addresses in the proximity of the requested address, and additionally recently accessed data will be accesses again (temporal and spacial locality). Prediction since its predicting which addresses will be accesses again.

4. Consider a 64-bit architecture computer executing the following code segment. How can this code be sped up using one of the "bit-level" parallelism techniques you learnt in class? **Please justify your answer.**

```
uint16_t a[100] = {0} // 16 bit integers initialized to 0
for (i = 0; i < 100; i++) {
     if (i > 50) {
          a[i] = 1;
     } else {
          a[i] += b[i]; // b is the same type and share as 'a'
and was initialized elsewhere in the program
     }
}
```

Hint: Note that up to 4 elements can be packed into a register

Can pack 4 uint16_t values in a single register (Packed word SIMD). The first loop will be a packed assignment, while the second loop can be a packed word addition

Since all elements of A have been initialized to zero, there is no scope of overflow

Question 1

You have to compare two computers each of which implements a different ISA. For a certain benchmark, computer One, which implements ISA-A, operates at 600 MHz with an IPC of 4. Computer two, which implements ISA-B, operates at 800 MHz with an IPC of 3.

(a) What is the performance in MIPS (Millions of Instructions Per Second) of the two machines for this benchmark?

(b) Which machine has better performance? Why?

(a)

IPC = Instructions per Cycle

Machine One:

Number of Instruction executed per second = IPC * Freq = 4 * 600 * 10^6 = 2400 MIPS

Machine Two:

Number of Instructions executed per second = IPC * Freq = 3 * 800 * 10^6 = 2400 MIPS

(b) Can't really compare the machines since the ISAs are different. Different ISAs' imply that we don't know the difference in Instruction counts between the two processors for the same benchmark.

Question 2

Suppose we have two unpipelined machines A and B, with operating frequencies 2GHz and 2.4GHz, respectively. Both machines contain only four types of instructions: ADD, MUL, BR, and MEM. Given the delay (i.e. cycles) for each operation on the machines, and the breakdown of benchmark programs X, Y and Z by instruction type, which machine (A or B) should you buy if you care equally about the performance of all 3 programs on it? **Please show your work including the speedup of the faster machine.**

	ADD	MUL	BR	MEM
Machine A	1	3	1	4
Machine B	1	4	2	2
Program X	30%	20%	10%	40%
Program Y	20%	40%	20%	20%
Program Z	25%	25%	20%	30%

Program X:
Let total number of instructions be x
Time taken by machine A:
(0.3*x + 0.2*3*x + 0.1*x + 0.4*4*x)/2 (ns) = 1.3*x (ns)

Time taken by machine B:
(0.3*1 + 0.2*4 + 0.1*2 + 0.4*2)/2.4 (ns) = 0.875*x (ns)

Speedup of B over A = 1.3*x/0.875*x = 1.48

Program Y:

Let total number of instructions be y
Time taken by machine A:
(0.2*1 + 0.4*3 + 0.2*1 + 0.2*4)*y/2 ns = 2.4*y/2 ns = 1.2*y ns

Time taken by machine B:
(0.2*1 + 0.4*4 + 0.2*2 + 0.2*2)*y/2.4 ns = 2.6*y/2.4 = 1.08*y ns

Speedup of B over A = 1.2*y/1.08*y = 1.11

Program Z:
Let total number of instructions be z
Time taken by machine A:
(0.25*1 + 0.25*3 + 0.2*1 + 0.3*4)*z/2 ns = 2.4*z/2 ns = 1.2*z ns

Time taken by machine B:
(0.25*1 + 0.25*4 + 0.2*2 + 0.3*2)*z/2.4 ns = 2.25*z/2.4 = 0.93*z ns

Speedup of B over A = 1.2/0.93 = 1.29

B is faster than A for all benchmarks. Hence B is better

Question 3

We have two benchmarks, X and Y. Each benchmark is divided into multiple subtasks each with the same number of operations. The number of operations in each subtask for different benchmarks is not necessarily the same. For benchmark X, machine A can process 2 subtasks per second, and machine B can process 3 subtasks per second. For benchmark Y, machine A can process 5 subtasks per second, and machine B can process 3 subtasks per second. If the weights of benchmark X and Y are equal, which machine would you choose? Please explain why.

Benchmark X:

Machine A:

- ⇨ 2 subtask in 1 s

  Time taken to execute one subtask = 0.5 s

Machine B:

- ⇨ 3 subtasks in 1 s

Time taken to execute one subtask = 1/3

Benchmark Y:

Machine A:

- ⇨ 5 subtask in 1 s

Time taken to execute one subtask = 1/5 s

Machine B:

- ⇨ 3 subtask in 1 s

Time taken to execute one subtask = 1/3 s

Average time taken by A to execute one subtask = (0.5 + 0.2)/2 = 0.35 s

Average time taken by B to execute one subtask = (1/3 + 1/3)/2 = 0.33 s

Execution time per subtask for B is less than A. B should be chosen over A.

(Note – Can also use harmonic mean directly on the subtask processing rates)

Question 4

Suppose we have a multiplication-intensive application X, and 20% of the executing time is spent by the multiplier. A smart architect optimizes the multiplier and improves the multiplication speed by 80% (1.8x).
1. What is the speedup of application X when this optimized multiplier is added?
2. If you are somehow able to further improve the multiplier's performance, what is the maximum possible speedup of X?

1. Applying Amdahl's law:

Speedup = 1/ ((1-0.2) + 0.2/1.8) = 1/(0.8 + 0.11) = 1.098

2. Let n be the speedup for multiplier.

Overall speedup : 1/ ((1-0.2) + 0.2/n)

Maximum possible speedup is possible when n tends to large number

As n-> infinity => 0.2/n -> 0

Speedup = 1/0.8 = 1.25